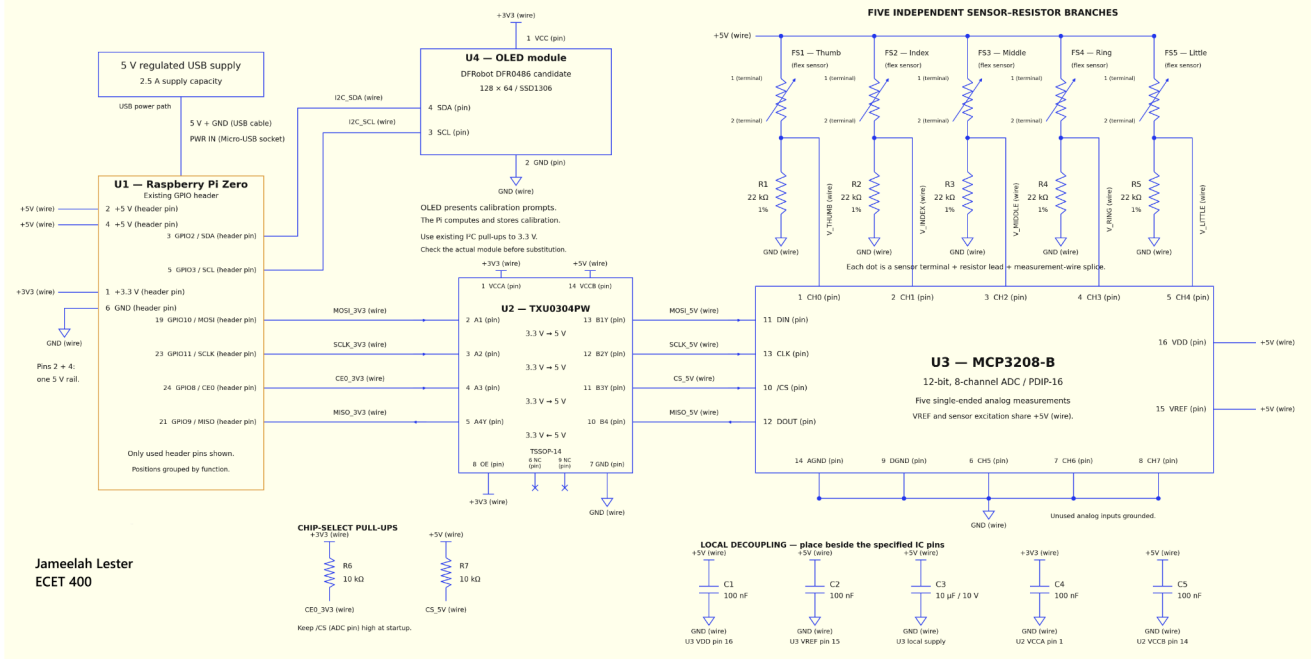

5 V regulated USB supply
2.5 A supply capacity
USB power path
5 V + GND (USB cable)
PWR IN (Micro-USB socket)
+3V3 (wire)
1 VCC (pin)
U4 — OLED module
DFRobot DFR0486 candidate
128 × 64 / SSD1306
I2C_SDA (wire)
I2C_SCL (wire)
4 SDA (pin)
3 SCL (pin)
2 GND (pin)
GND (wire)
OLED presents calibration prompts.
The Pi computes and stores calibration.
Use existing I²C pull-ups to 3.3 V.
Check the actual module before substitution.
U1 — Raspberry Pi Zero
Existing GPIO header
+5V (wire)
2 +5 V (header pin)
+5V (wire)
4 +5 V (header pin)
3 GPIO2 / SDA (header pin)
5 GPIO3 / SCL (header pin)
+3V3 (wire)
1 +3.3 V (header pin)
6 GND (header pin)
GND (wire)
Pins 2 + 4:
one 5 V rail.
19 GPIO10 / MOSI (header pin)
23 GPIO11 / SCLK (header pin)
24 GPIO8 / CE0 (header pin)
21 GPIO9 / MISO (header pin)
Only used header pins shown.
Positions grouped by function.
MOSI_3V3 (wire)
SCLK_3V3 (wire)
CE0_3V3 (wire)
MISO_3V3 (wire)
+3V3 (wire)
1 VCCA (pin)
+5V (wire)
14 VCCB (pin)
U2 — TXU0304PW
2 A1 (pin)
3 A2 (pin)
4 A3 (pin)
5 A4Y (pin)
3.3 V → 5 V
3.3 V → 5 V
3.3 V → 5 V
3.3 V ← 5 V
TSSOP-14
6 NC (pin)
9 NC (pin)
8 OE (pin)
7 GND (pin)
+3V3 (wire)
GND (wire)
13 B1Y (pin)
12 B2Y (pin)
11 B3Y (pin)
10 B4 (pin)
MOSI_5V (wire)
SCLK_5V (wire)
CS_5V (wire)
MISO_5V (wire)
FIVE INDEPENDENT SENSOR-RESISTOR BRANCHES
+5V (wire)
FS1 — Thumb
(flex sensor)
FS2 — Index
(flex sensor)
FS3 — Middle
(flex sensor)
FS4 — Ring
(flex sensor)
FS5 — Little
(flex sensor)
1 (terminal)
2 (terminal)
R1
22 kΩ
1%
R2
22 kΩ
1%
R3
22 kΩ
1%
R4
22 kΩ
1%
R5
22 kΩ
1%
V_THUMB (wire)
V_INDEX (wire)
V_MIDDLE (wire)
V_RING (wire)
V_LITTLE (wire)
GND (wire)
Each dot is a sensor terminal + resistor lead + measurement-wire splice.
U3 — MCP3208-B
12-bit, 8-channel ADC / PDIP-16
Five single-ended analog measurements
VREF and sensor excitation share +5V (wire).
1 CH0 (pin)
2 CH1 (pin)
3 CH2 (pin)
4 CH3 (pin)
5 CH4 (pin)
11 DIN (pin)
13 CLK (pin)
10 /CS (pin)
12 DOUT (pin)
16 VDD (pin)
15 VREF (pin)
+5V (wire)
+5V (wire)
14 AGND (pin)
9 DGND (pin)
6 CH5 (pin)
7 CH6 (pin)
8 CH7 (pin)
GND (wire)
Unused analog inputs grounded.
CHIP-SELECT PULL-UPS
+3V3 (wire)
R6
10 kΩ
CE0_3V3 (wire)
+5V (wire)
R7
10 kΩ
CS_5V (wire)
Keep /CS (ADC pin) high at startup.
LOCAL DECOUPLING — place beside the specified IC pins
+5V (wire)
C1
100 nF
GND (wire)
U3 VDD pin 16
+5V (wire)
C2
100 nF
GND (wire)
U3 VREF pin 15
+5V (wire)
C3
10 µF / 10 V
GND (wire)
U3 local supply
+3V3 (wire)
C4
100 nF
GND (wire)
U2 VCCA pin 1
+5V (wire)
C5
100 nF
GND (wire)
U2 VCCB pin 14
Jameelah Lester
ECET 400